# IMO 2026 — Autonomous Model Comparison

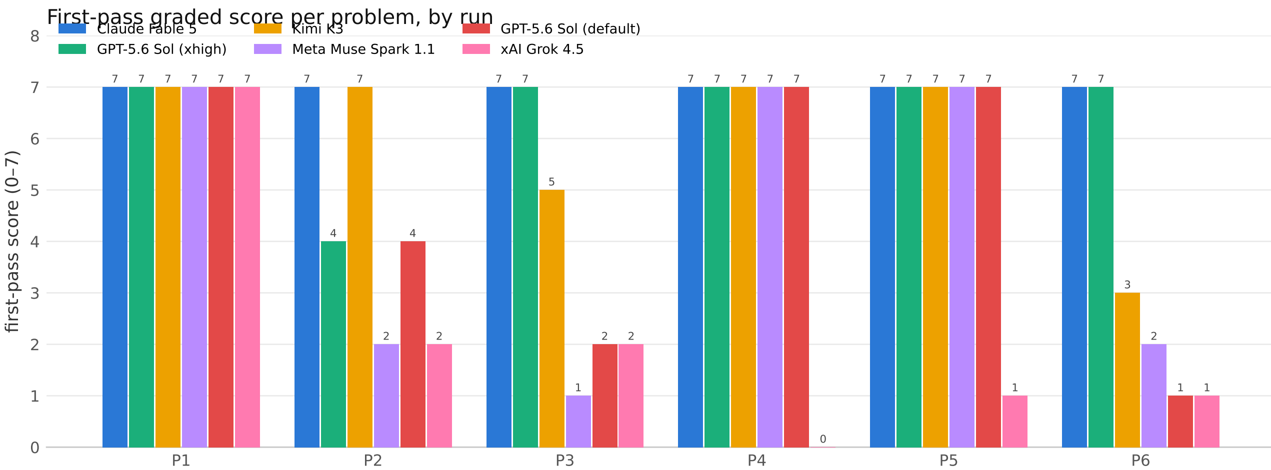
Independently graded results across six runs (five models)

One identical minimal agent harness for every run · IMO 0–7 grading by independent verifier agents, claims checked symbolically and by machine

## Final graded scores

| Run | P1 | P2 | P3 | P4 | P5 | P6 | **Total** |
|---|---|---|---|---|---|---|---|
| Claude Fable 5 (default `high`) | 7 | 7 | 7 | 7 | 7 | 7 | **42/42** |
| GPT-5.6 Sol (`xhigh`)[†] | 7 | 7 | 7 | 7 | 7 | 7 | **42/42** |
| Kimi K3 (default)[†] | 7 | 7 | 7 | 7 | 7 | 7 | **42/42** |
| GPT-5.6 Sol (default) | 7 | 4 | 2 | 7 | 7 | 1 | **28/42** |
| Meta Muse Spark 1.1 (`xhigh`) | 7 | 2 | 1 | 7 | 7 | 2 | **26/42** |
| xAI Grok 4.5 (`xhigh`)[‡] | 7 | 2 | 2 | 0 | 1 | 1 | **13/42** |

[†] reached via repair rounds seeded with reviewer defect findings (no external solutions). [‡] Grok needed a harness guard to force it to actually call the write-tool (it repeatedly narrated writing files without doing so). Claude Fable 5 is the only run graded 42/42 on its first pass; Muse Spark 1.1 and Grok 4.5 ran single-pass, so their scores are both first-pass and final.




## Time and cost (all rounds included)

| Run | Final | Total time | Cost (USD) | Output tokens |
|---|---|---|---|---|
| Claude Fable 5 | 42/42 | 2.5 h | $51.05 | 705k |
| GPT-5.6 Sol (`xhigh`) | 42/42 | 3.8 h | ∼$20.54 | 236k |
| Kimi K3 | 42/42 | 17.4 h | ∼$31.40 | 1,545k |
| Meta Muse Spark 1.1 | 26/42 | 2.3 h | ∼$4.45 | 761k |
| GPT-5.6 Sol (default) | 28/42 | 1.0 h | ∼$4.04 | 80k |
| xAI Grok 4.5 | 13/42 | 3.2 h | ∼$14.60 | 606k |

Anthropic API metering for Claude Fable 5; OpenRouter list pricing for GPT-5.6 Sol, Muse Spark 1.1, Grok 4.5; Kimi K3 estimated from logged tokens at list pricing. Times include every retry and repair round.

## Findings

- **A frontier tier and a challenger tier.** Claude Fable 5, GPT-5.6 Sol, and Kimi K3 all reach 42/42. The two newest models land far below on first pass — Meta Muse Spark 1.1 at 26/42, xAI Grok 4.5 at 13/42 — both solving the routine problems but neither closing the geometry (P2), combinatorics capstone (P3), or number-theory finale (P6).
- **Claude Fable 5 was perfect first-pass**, at its default effort, with no reviewer and no second chances.
- **Effort and feedback lift the frontier models.** GPT-5.6 Sol went 28→39/42 from default to `xhigh`; reviewer-defect feedback then brought GPT (`xhigh`) and Kimi K3 to 42/42. The same levers did not rescue the challengers, whose failures were deeper (wrong answers, unproven cruxes) than the frontier models' final blemishes.
- **Muse got P3 wrong** (conjectured $c = (n+1)/(2n+1)$, false for all $n \geq 2$, confirmed by exact minimax). **Grok has a tool-adherence failure**: on P4 and P5 it ended turns claiming to write files while making zero tool calls, producing nothing until a harness guard forced a real write.
- **Self-reports inflate.** Almost every run self-reported every problem "solved"; independent grading confirmed far fewer, each deduction backed by a named, checkable defect.

## Caveats

- Graders are Claude-based agents, not human medalists.
- All six runs share the identical harness and are directly comparable; reasoning-effort settings are noted per run.
- Full per-turn audit trails (every round of every run) and per-problem grader verdicts with named defects accompany this document; see REPORT.md for attempts/time/tokens/cost.